

Received 18 February 2025, accepted 12 March 2025, date of publication 19 March 2025, date of current version 28 March 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3552934

RESEARCH ARTICLE

Conformal Prediction for Uncertainty Quantification and Reliable HER2 Status Classification in Breast Cancer IHC Images

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This work was supported by Thailand Program Management Unit (PMU-B) Grant for Multi-Institutional AI Development in Digital Pathology under Grant B05F640233.

ABSTRACT Accurate assessment of Human Epidermal growth factor receptor 2 (HER2) status in breast cancer is crucial for determining treatment eligibility for targeted therapies. Although machine learning approaches have shown promise in automating HER2 status classification from immunohistochemistry (IHC) images, they struggle to maintain reliable accuracy in equivocal cases. This study introduces a novel application of Conformal Prediction (CP) framework that, instead of making potentially incorrect predictions for uncertain cases, explicitly identifies them for further examination. The CP framework achieves this by generating prediction sets whose size reflects uncertainty, inherently aligning with the clinical practice of identifying cases that require confirmatory testing. Using dual in situ hybridization (DISH) results as the gold standard for HER2 status, we developed our system using handcrafted features—including color intensity, local binary patterns, and Haralick features—combined with tree-based classifiers. In experiments on 2,570 IHC images (94 cases) from the Institute of Pathology, Bangkok, Thailand, our framework demonstrated the ability to make highly accurate predictions on unambiguous cases while automatically filtering out uncertain ones. At a significance level of 0.05, the XGBoost-based conformal predictor achieved 90.1% (95% CI: 87.4–92.2%) accuracy on the equivocal cases, while identifying 69.8% (95% CI: 67.7–71.9%) of cases as requiring additional testing. This selective prediction approach represents a practical step toward reliable automated HER2 assessment by focusing on high-confidence predictions. The source code of our implementation is available at: <https://github.com/Surayuth/her2cp.git>.

INDEX TERMS Breast cancer, conformal prediction, digital pathology, handcrafted features, HER2 status, immunohistochemistry, uncertainty quantification.

I. INTRODUCTION

Breast cancer is the most common malignancy and the leading cause of death in women worldwide [1]. Early

The associate editor coordinating the review of this manuscript and approving it for publication was Wai-Keung Fung¹.

detection improves treatment outcomes and significantly increases the likelihood of survival [2]. One key factor contributing to breast cancer progression is amplification of the Human Epidermal growth factor receptor 2 (HER2) gene, which results in uncontrolled cell growth. Therefore, testing for HER2 status is crucial to determine the appropriate

treatment strategy, particularly for HER2-positive cancers that benefit from targeted therapy.

Two widely used methods for assessing HER2 status are immunohistochemistry (IHC) and in situ hybridization (ISH) [3], [4]. Although ISH test is the gold standard for detecting HER2 amplification, it is expensive and time-consuming. Consequently, the first screening method employs the IHC test owing to its affordability, speed, and broad availability. IHC uses 3,3'-diaminobenzidine (DAB) staining to produce a brown coloration in areas where the HER2 protein is present.

IHC scores range from 0 to 3+ according to the American Society of Clinical Oncology/College of American Pathologists (ASCO/CAP) guidelines [3], with scores of 0 and 1+ indicating HER2-negative, and a score of 3+ indicating HER2-positive. A score of 2+ requires further ISH tests to accurately determine the HER2 status. These guidelines evaluate three key aspects: the percentage of tumor cells exhibiting membrane staining, staining intensity, and completeness of membrane staining. A score of 3+ is assigned when more than 10% of the tumor cells show robust and complete membrane staining. A score of 1+ indicates incomplete, faint membrane staining in more than 10% of tumor cells, while a score of 0 is assigned when no staining or incomplete membrane staining is observed in less than 10% of tumor cells. Equivocal cases (2+) are characterized by weak to moderate complete membrane staining in 10% or more of tumor cells. The difficulty in assigning scores to samples that fall near the thresholds between categories (such as between 1+ and 2+, or between 2+ and 3+) can lead to variability in interpretation among pathologists. Fig. 1 shows representative regions for each IHC score.

A significant limitation of visual IHC scoring is the subjectivity of pathologists' evaluations, which often leads to the referral of many equivocal cases for ISH testing. Digital image analysis (DIA) software offers an alternative by assessing the connectivity of DAB staining, achieving high concordance with ISH test results [5], [6], which results in the reduction of equivocal cases. Although these techniques have been developed to improve diagnostic consistency, they remain largely inaccessible due to their high cost and limited availability in many diagnostic centers. Additionally, the proprietary nature of commercial DIA software obscures their feature extraction algorithms, hindering the transparency and independent validation of their analytical methods. Moreover, studies by Koopman et al. [7] showed that a significant reduction in equivocal cases can lead to loss of sensitivity. Therefore, further analysis is required to evaluate both the trade-off between overall performance and the reduction in equivocal cases, as well as the interpretability and transparency of the analytical methods employed.

Recent advances have leveraged machine learning (ML) and deep learning (DL) models to predict IHC scores and HER2 status. ML approaches typically rely on handcrafted features that are directly extracted from images. The

DAB-stained region can be segmented by determining the appropriate color channel using color deconvolution [8] or color transformation [9], followed by automatic color thresholding. Handcrafted features are generally categorized as color and texture features, which have been used in combination in previous studies. Local binary patterns, Haralick features, and stain connectivity are the most commonly used texture features in recent studies. Mukundan [10] used multi-level thresholding to extract the characteristics of local binary patterns to predict IHC scores. Aguilera et al. [11] trained and ensembled machine learning models using color and Haralick features. Tewary et al. [12] applied image processing and skeletonization to extract the ratio of the complete membranes and measured their connectivity. An alternative approach is based on DL models, where convolutional neural networks (CNNs) automatically learn abstract features from IHC images. Early CNN-based studies relied on supervised training, in which pathologists manually selected images for training [13], [14], [15], [16]. However, recent work has adopted unsupervised learning methods, allowing the model to automatically sample stained regions for training [17], [18], [19]. Despite the high accuracy of both ML and DL approaches in predicting IHC scores, there is still a need to analyze their performance in determining HER2 status, particularly in equivocal cases.

To address the inherent uncertainty in predicting HER2 status from IHC images, we employed Conformal Prediction (CP) [20], [21], a framework that generates prediction sets with a user-specified error rate. CP provides flexibility by categorizing predictions into three possible outcomes: positive, negative, and ambiguous. Ambiguous predictions, often arising in equivocal cases, are flagged for further confirmatory testing (e.g., FISH or DISH) to ensure accurate HER2 status determination. Furthermore, CP allows for a trade-off between prediction certainty and the number of ambiguous cases, making it helpful in prioritizing accurate predictions, even if additional testing is required.

The primary focus of this study was to explore how error rate is related to confidence in the prediction of HER2 status. Our research can help identify the benefits of reducing the number of equivocal cases and the risk of misidentifying the true HER2 status when deploying a machine learning model for hospitals. Owing to the limited dataset and high variability in illumination conditions, we opted for a traditional machine-learning model instead of a CNN. This choice helps to mitigate overfitting and allows for more efficient training. These models can also perform inference in real-time without requiring a graphics processing unit (GPU), better reflecting real-world use cases in hospital settings. We selected color intensity, local binary patterns, and Haralick features as handcrafted features in our experiments owing to their high predictive power for IHC score in previous studies. The machine learning approach also requires fewer computational resources than the deep learning approach.

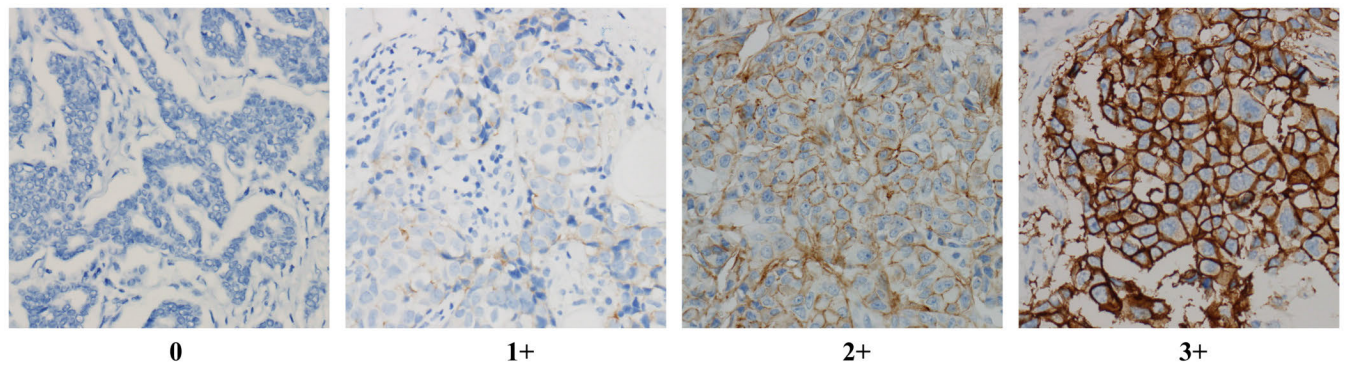


FIGURE 1. Sampled stained regions from images with IHC scores 0 to 3+.

The contributions of this work are:

- 1) We applied Conformal Prediction in HER2 status prediction for uncertainty quantification.
- 2) We analyzed the impact of the significance level on the miscoverage rate, ambiguity rate, and accuracy of predictions.
- 3) We compared the performance of CP on multiple tree-based models.

II. RELATED WORK

Extensive research exists on uncertainty quantification in histopathological image classification of HER2 status. However, most previous studies have used deep learning models as classifiers, making their approach model-dependent. For example, Monte Carlo dropout was applied to estimate the uncertainty and confidence by randomly dropping out neural nodes, which generated a distribution of predictions for each input sample [22], [23]. The deep ensemble approach used multiple neural networks initialized differently to estimate the uncertainty of predictions [24], [25].

Some model-agnostic approaches, such as Test-Time Augmentation (TTA), have been reported to yield performances similar to those of the deep ensemble method [24]. However, TTA requires higher computational time because multiple transformations are required to generate the uncertainty distribution. Although probability calibration techniques, such as Platt scaling [26], can be used to approximate the confidence of prediction, they do not inherently provide the uncertainty region for handling equivocal cases.

Based on the aforementioned limitations, we explored a distribution-free uncertainty quantification framework called Conformal Prediction (CP) [20], [21]. CP generates a prediction set instead of a point prediction, and the prediction set is expected to marginally miss a predefined percentage of true classes. We refer to this predefined miscoverage rate or error rate as the significance level. In binary classification, the prediction set may include one, both, or none of the classes. A class is included in the prediction set based on how likely it is to be a true class. A *nonconformity measure* is defined to quantify the likelihood, which is

referred to as the nonconformity score in the CP. There are countless definitions of the nonconformity measure, some of which are model-agnostic, while others are not. For example, the similarity between the terminal nodes from the Random Forest classifier of two samples was used as the nonconformity score [27], [28]. Model-agnostic measures rely on probability scores to quantify the similarity. Previous studies [29], [30] empirically showed that the size of the prediction set was highly dependent on the selected measure.

Although research exists on applying CP to various histopathology datasets [25], [31], [32], there has been no work on using CP to estimate uncertainty for both IHC score and HER2 status.

III. MATERIAL AND METHODS

A. IMAGE ACQUISITION

Cases included in the study were retrieved from the Institute of Pathology, Department of Medical Services, Ministry of Public Health, Bangkok, Thailand. All HER2 IHC images were captured by pathologists either by microscope equipped with digital camera or by image capturing from the digitalized scanned HER2 IHC. The ethical approval for this study was granted by the Institute of Pathology, under the project entitled “Multi-Institutional AI Development in Digital Pathology” (Project #IOP-KM_R65-001).

The IHC score of each case was assigned according to the ASCO/CAP guidelines. DISH tests were performed on formalin-fixed, paraffin-embedded (FFPE) tissue specimens of invasive carcinoma to ensure proper preservation for microscopic examination. Tissue sections were stained using the Ventana Benchmark XT system (Roche Diagnostics) with an Inform HER2 Dual ISH DNA Probe Cocktail (Ventana Medical Systems). DNA probes targeting HER2 and CEP17 were used, where HER2 signals appeared as black dots and CEP17 signals appeared as red dots under microscopic examination. The HER2 and CEP17 signals were visualized using UltraView SISH DNP and UltraView Red ISH DIG Detection kits, respectively. The DISH status was evaluated according to the ASCO/CAP guidelines, which defined HER2 amplification as positive when the HER2/CEP17 ratio

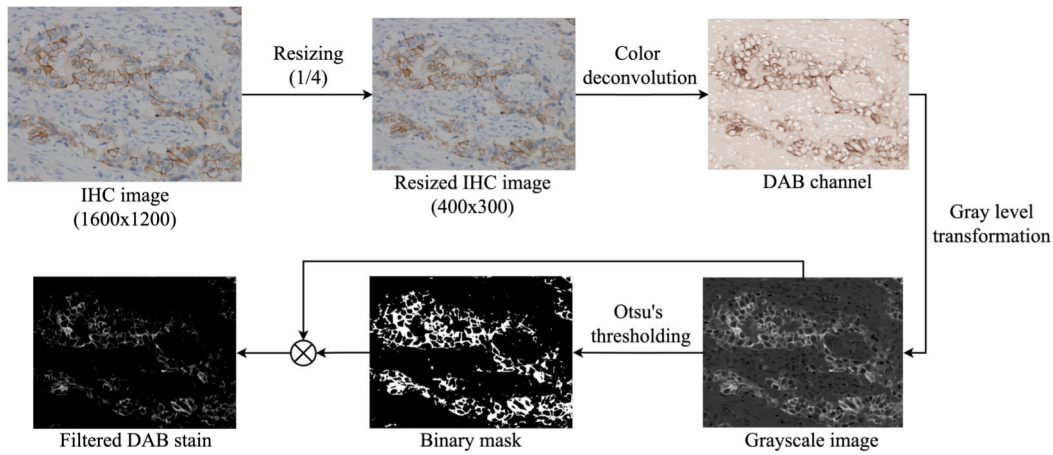


FIGURE 2. Image preprocessing pipeline for DAB stain extraction from IHC images. The $\otimes$ symbol denotes point-wise multiplication between the binary mask and grayscale image.

is ≥ 2.0 or when the average HER2 signals per cell is ≥ 6.0 . Because of its ability to directly visualize and quantify HER2 gene copy numbers, DISH served as the gold standard in this study.

Table 1 summarizes the number of images and cases used in the experiments based on the DISH status. The resolution of the captured IHC images was $1,600 \times 1,200$ pixels, corresponding to a tissue area of approximately 400×300 micrometers (0.12 mm^2) at 400x magnification. For each case, 10-30 images were manually selected to balance statistical rigor, tissue representation, and practical workflow constraints. This ensured sufficient coverage for each case while managing the workload of the pathologist.

For model development and evaluation, nested cross-validation was performed by repeatedly splitting the dataset into training, testing, and calibration sets with different random seeds. Stratification was based on IHC scores (0, 1+, 2- with DISH-, 2+ with DISH+, 3+), which inherently maintained the distribution of DISH status in the splits. Across all splits, the dataset distribution averaged approximately 1,534 images (56 cases) for training, 512 images (19 cases) for testing, and 525 images (19 cases) for calibration. This cross-validation approach, rather than a single predetermined split, ensures more a robust evaluation of the model performance across different data distributions while maintaining independent test and calibration sets.

B. FEATURE EXTRACTION

1) IMAGE PREPROCESSING

This study used 19 distinct features: the DAB-stained region ratio, mean DAB intensity, ten local binary patterns, and seven Haralick texture features. Fig.2 illustrates the preprocessing pipeline. To balance computational efficiency and detail preservation, we resized the images to 1/4 of their original dimensions (from $1,600 \times 1,200$ to 400×300 pixels). We employed color deconvolution [8] to separate the IHC image into its constituent DAB and hematoxylin

TABLE 1. Summary of number of cases and IHC images based on DISH status used in this work.

DISH status	IHC score	Cases	Images
Negative	0	11	324
	1+	11	266
	2+	23	617
Positive	2+	30	833
	3+	19	530
Total		94	2,570

components. The DAB channel was converted into grayscale intensity values for further analysis. Then, we applied Otsu's thresholding [33] to segment the stained regions from the background in the grayscale image. From these segmented regions, we computed two features: the stained region ratio and mean DAB intensity. The subsequent sections detail the extraction of the local binary patterns and Haralick features from the processed images.

2) LOCAL BINARY PATTERN

Local Binary Patterns (LBPs) [34], [35] are texture descriptors that assign a binary number to each pixel p based on the intensity differences between pixel p and its neighboring pixels I_n . LBPs capture the local texture characteristics of tissue and relative color intensity, which are more robust to distribution shifts than the absolute intensity. This robustness is particularly important in IHC images, where variability in brightness and illumination can be high. Let N be the number of neighboring pixels of pixel p ; Each digit of the LBP is determined by the following rule:

$$d_n = \begin{cases} 0, & \text{if } I_p \leq I_n \\ 1, & \text{if } I_p > I_n \end{cases} \quad (1)$$

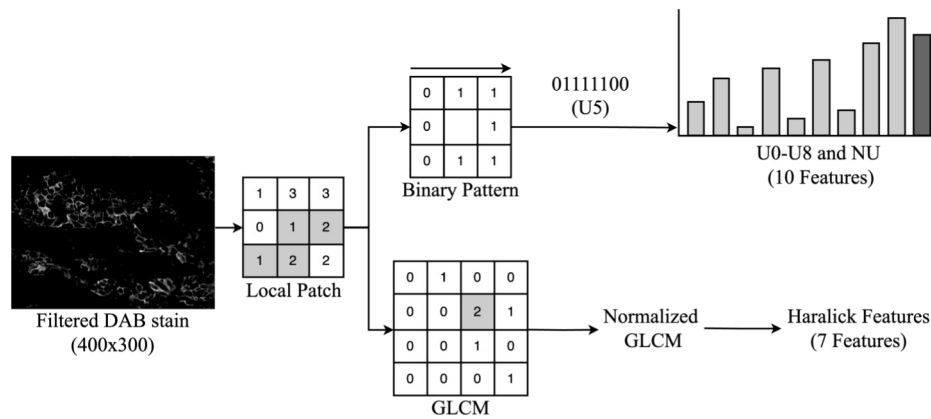


FIGURE 3. Feature extraction process for LBP and Haralick features. The local 3×3 patch is processed in two parallel paths: (1) LBP computation compares each pixel with its center pixel to generate a binary pattern, which is read clockwise starting from the top-left corner to form pattern U5=01111100. The pattern frequencies are accumulated into a histogram with nine uniform patterns (U0-U8) and one non-uniform pattern (NU). (2) GLCM computation counts the frequency of intensity value pairs in the local patch (shaded cells illustrate example pairs) to form a 4×4 co-occurrence matrix. Seven Haralick features are then computed from the normalized GLCM.

To ensure consistency, it is essential to assign a pattern to one of the neighboring pixels and maintain a consistent direction, either clockwise or counter-clockwise. A pattern is considered uniform if it contains at most two bit-wise transitions when traversed circularly; otherwise, it is deemed non-uniform. For $N = 8$, there are 256 possible patterns, which can be grouped into 58 uniform patterns and one non-uniform pattern. A histogram of these 59 features is typically used for image texture classification.

However, the histogram of uniform patterns is not inherently rotation-invariant, leading to increased feature variation when the image is rotated. This variation can hinder the ability of the model to generalize across different orientations of tissue samples, which may vary during image acquisition. To address this issue, we grouped uniform patterns containing the same number of 1s, reducing the size of the uniform histogram to nine bins. All non-uniform patterns were grouped into a single bin, resulting in a total of ten features extracted from the LBP histogram. This approach achieves rotation invariance, thereby ensuring that the model remains consistent despite changes in sample orientation.

An example of an LBP calculation is shown in Fig. 3. The uniform pattern U5 from the figure is assigned to the uniform histogram, which is then normalized.

3) HARALICK FEATURES

Haralick features [36] are derived from the Gray Level Co-occurrence Matrix (GLCM), which is a statistical tool used to analyze texture by examining the spatial relationships between pixel intensities in an image. These features are well-suited for this study as they effectively capture texture details indicative of different tissue characteristics, making them valuable for distinguishing between HER2 statuses.

The GLCM represents texture information in a square matrix format. For a grayscale image with N intensity levels,

this matrix has dimensions $N \times N$. Each element (i, j) in the matrix represents how frequently two pixels with intensities i and j appear together at a specific distance and orientation, as illustrated in Fig. 3. The spatial relationship between pixel pairs is defined by a polar vector (r, θ) , where r is the radial distance and θ represents the direction. We considered four directions at 90° intervals: 45° , 135° , 225° , and 315° . To improve the computational efficiency while maintaining sufficient detail for texture analysis, we quantized the pixel intensities inside the segmented regions to 16 levels.

Seven Haralick features were selected for this study: contrast, dissimilarity, homogeneity, Angular Second Moment (ASM), energy, entropy, and correlation. Both ASM and Energy were included in the experiments as their relations are not linear, ensuring that the information they provide is not redundant for tree-based models. Appendix A provides detailed definitions of each feature.

C. CONFORMAL PREDICTION

Conformal prediction operates as a post-processing framework on pre-trained models, providing statistical guarantees that predictions will include the true class with a probability of $(1 - \text{significance level})$. For example, at a significance level of 0.05, CP guarantees that 95% of the prediction sets contain the true class. Unlike conventional classification where a probability threshold (e.g., 0.95) does not guarantee that 95% of predictions contain true classes, CP ensures reliable coverage through these statistical guarantees.

CP can be categorized into Transductive Conformal Prediction (TCP) and Inductive Conformal Prediction (ICP) [37]. These approaches differ in how the training and testing datasets are used to construct the prediction set. ICP requires the model to be trained only once before the test step. The disadvantage of this method is that a portion of the

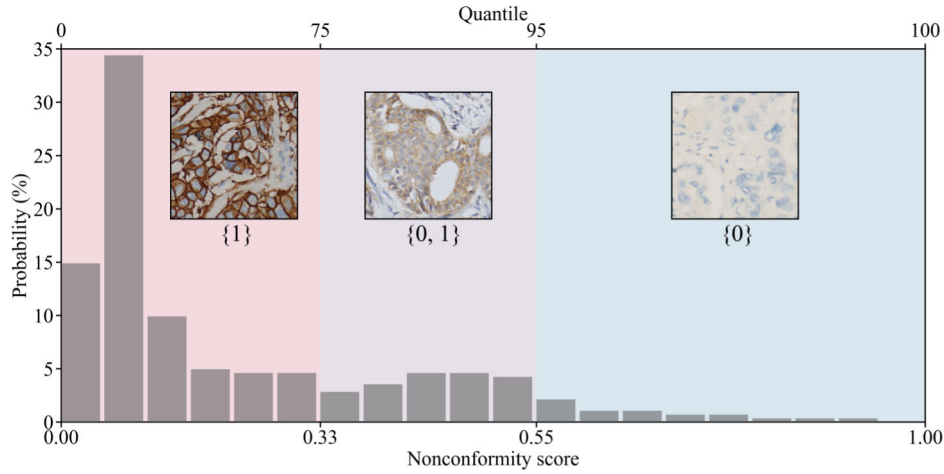


FIGURE 4. Distribution of nonconformity scores for the positive class computed from the calibration set at a significance level 0.05. The positive class (1) is included in the prediction set for a test image if its nonconformity score for the positive class is below 0.55 (Q95). The distribution exhibits three distinct regions based on nonconformity scores. In the pink region, low scores typically yield confident singleton predictions of 1, corresponding to images with strong staining patterns (left). In the gray region near Q95, samples tend to include both classes in the prediction set, reflecting prediction uncertainty, and are associated with equivocal images (middle). In the blue region, high scores typically yield confident singleton predictions of 0, as the nonconformity score for the negative class is very low, corresponding to images with weak staining patterns (right).

data must be reserved as a calibration set to construct reference nonconformity scores. TCP, in contrast, uses the entire dataset, including the test sample, to train a new model for each test prediction set. Compared to ICP, this approach generates smaller prediction sets because more data are available for reference [38]. Despite TCP's more efficient use of data, we implemented ICP in our study because its computational efficiency and real-time applicability.

Using this approach, we applied ICP for uncertainty quantification over a trained machine learning model $\hat{f}$ as follows:

- 1) Define the nonconformity measure: This function requires a trained model $\hat{f}$, input features $\mathbf{x}$, and its class y to produce the nonconformity score. In this study, we chose a model-agnostic nonconformity measure called *hinge* [29], which is defined as

$$s_y(\mathbf{x}) = 1 - \hat{f}(\mathbf{x})_y. \quad (2)$$

Using this definition, only the true class was used to calculate the nonconformity score. Thus, a prediction with high confidence in the true class yields a small nonconformity score.

- 2) Construct nonconformity score distributions: Nonconformity scores were computed using all the samples from the calibration set. This process was performed separately for each class. For the binary classification, we obtained two distributions.
- 3) Calculate a separable quantile: The distributions constructed in the previous step informed us where the true class would likely be. In this step, we calculated an empirical quantile of the nonconformity scores for each

class c using the calibration samples n_c

$$\hat{q}_c(\alpha) = q\left(s_c, \frac{[(n_c + 1)(1 - \alpha)]}{n_c}\right) \quad (3)$$

such that the expected miscoverage rate of each class is equal to the predefined significance level α . CP relies on the exchangeability assumption between the test data and calibration set for a valid coverage guarantee.

- 4) Testing: We computed the nonconformity score for each class of test sample $\hat{\mathbf{x}}$ according to (2). The prediction set Γ was constructed by including all classes in which the nonconformity score was less than or equal to (3):

$$\Gamma(\hat{\mathbf{x}}, \alpha) = \{c | s_c(\hat{\mathbf{x}}) \leq \hat{q}_c(\alpha), c \in \{0, 1\}\}. \quad (4)$$

Fig. 4 illustrates the construction of the prediction sets based on nonconformity scores. The figure shows the distribution of the nonconformity scores for the positive class, highlighting the thresholds and regions that determine the inclusion of classes in the prediction sets.

D. EVALUATION METRICS

In this study, we considered three metrics: miscoverage rate, ambiguity rate, and accuracy. All metrics were calculated at both image and case levels. For the case-level predictions, we first predicted the status of each image in the case as negative, positive, or ambiguous. We then aggregated these image-level predictions to determine the case-level status using majority rule. A status (negative, positive, or ambiguous) is assigned to a case in which more than half of its images have that status. Otherwise, the case is classified as ambiguous.

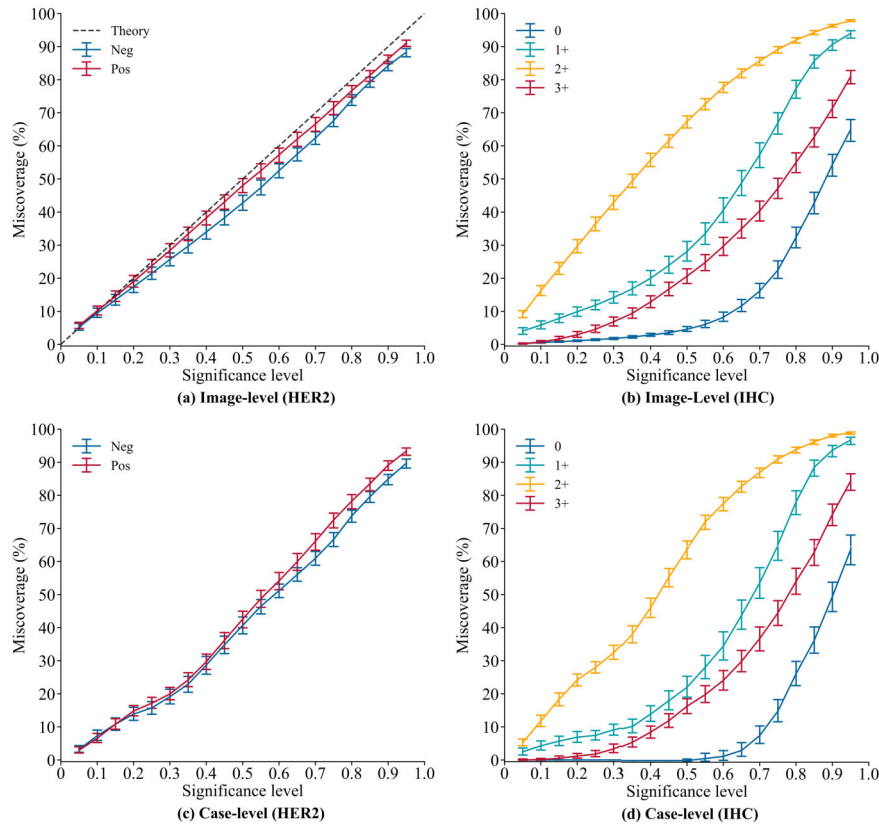


FIGURE 5. Miscoverage rates from Random Forest classifier. Image-level: HER2 status (a) and IHC score (b). Case-level: HER2 status (c) and IHC score (d). A trend line is shown only in (a), as calibration was performed at the image level.

TABLE 2. Criteria for case-Level aggregation.

Criteria	Satisfy	Pred. set	Status
Majority image-level class occupies over 50% with no ties	Y	$\emptyset$	Ambiguous
		{0}	Negative
		{1}	Positive
	N	{0, 1}	Ambiguous
		{0, 1}	Ambiguous

For instance, in a case with 100 images, if 55 images are positive, the case is classified as positive. However, if 30 images are negative, 40 are positive, and 30 are ambiguous, the case is classified as ambiguous since no status achieves the majority threshold. This choice of aggregation is used in this work to ensure that ambiguous predictions on case level are preferred rather than making predictions with less uncertainty. Clinically, it ensures that a patient gets the further test even though it results in more false positives.

Table 2 summarizes the aggregation method for case-level predictions from the image-level results. We chose the term “ambiguous” instead of “equivocal” to avoid potential misinterpretation, as these cases may not necessarily meet the strict criteria for equivocality defined by pathologists.

The metrics were evaluated using both the IHC scoring (0-3+) and binary HER2 status (positive/negative) systems. When calculating the metrics for specific groups, the denominator (total prediction set) varied based on the reference system and analysis level. For example, when evaluating metrics for equivocal predictions at the case level using the IHC system, the total prediction set comprised cases assigned as 2+ by the pathologists. Similarly, when calculating the metrics for HER2-positive predictions, the total prediction set included all cases classified as HER2-positive.

$$\text{Miscoverage rate} = \frac{\text{Number of sets without true class}}{\text{Total prediction sets}} \quad (5)$$

This metric represents the reliability of the CP framework, as it is expected to be linearly related to the user-defined significance level of a well-calibrated conformal predictor.

$$\text{Ambiguity rate} = \frac{\text{Sets with size} \neq 1}{\text{Total prediction sets}} \quad (6)$$

Although a lower ambiguity rate is preferable because it implies fewer ISH tests, correctly identifying HER2-positive cases is the primary priority.

$$\text{Accuracy} = \frac{\text{Correct predictions for HER2 status}}{\text{Total prediction sets}} \quad (7)$$

When calculating the accuracy for specific IHC groups, we measured the correct HER2 status predictions, not IHC scores, because our model only predicts HER2 status. For example, when calculating the accuracy for IHC 2+ cases, we measured how often the model correctly identified the HER2 status. This reflects our focus on clinical HER2 status rather than precise IHC scoring.

E. TRAINING AND VALIDATION

We selected tree-based models such as Decision Trees (DT), Random Forests (RF), Gradient-Boosting Trees (GBT), and Extreme Gradient Boosting Trees (XGB) as classifiers because they achieve optimal performance without requiring feature scaling. Additionally, their varying levels of complexity allowed us to examine the effect of model performance on conformal predictions. The details of the hyperparameter optimization for each model are provided in Appendix B.

We employed nested cross-validation (NCV) to train and test the model to ensure reliable performance estimation. A simple train-test split can introduce high variance. Although standard cross-validation (CV) reduces this variance, it may still overestimate the generalization error when tuning hyperparameters for the same data split. The NCV mitigates this by using an inner loop for hyperparameter tuning and an outer loop for performance evaluation, ensuring a more robust estimate of model generalizability. Compared with CV, NCV uses the validation set from each fold as a test set (outer fold) and performs training and validation processes on the training folds (inner folds). The value of each metric was estimated by averaging the results from all outer folds.

In our study, the cases were stratified into five groups based on the IHC score to ensure a balance between the HER2 status and IHC score. One of the five parts was reserved as a calibration set, whereas the remaining data were used for the NCV to estimate the performance.

We repeated NCV 100 times for each set of significance levels. These 100 values from each metric were then resampled with 10,000 replacements to estimate the 95% confidence intervals (CI) using the bias-corrected bootstrap. All experiments were conducted on Ubuntu 24.04 LTS with an Intel Core i7-14700K processor and 16GB RAM. Owing to variations in random number generation across different operating systems and computational environments, the exact numerical reproduction may vary slightly, while the demonstrated statistical trends and patterns remain robust and reproducible. The complete source code and implementation details are available at <https://github.com/Surayuth/her2cp.git>, containing all the scripts and documentation necessary to reproduce our results.

IV. EXPERIMENTS AND ANALYSIS

A. MISCOVERAGE RATE ANALYSIS

The image-level miscoverage results for HER2 status demonstrated a strong alignment with the theoretical expectations, as illustrated in Fig. 5(a). The marginal miscoverage

exhibited a linear increase with respect to the significance level. For both the positive and negative classes, the observed miscoverage rates remained consistently below theoretical expectations, validating the coverage guarantee of our conformal predictor.

Analysis of IHC score categories at the image level (Fig. 5(b)) revealed distinct prediction reliability patterns. Categories 0, 1+, and 3+ exhibited the lowest marginal miscoverage rates, indicating a superior prediction reliability for these classes. However, category 2+ demonstrated notably higher miscoverage rates, underscoring the inherent challenges in accurate HER2 status determination in equivocal cases. These findings corroborate clinical observations that equivocal cases traditionally present greater diagnostic complexity.

The case-level analysis (Fig. 5(c,d)) yielded similar trends, with miscoverage patterns mirroring the image-level results for both HER2 status and IHC scores. This consistency across the different analytical granularities substantiates the robustness of our methodological approach.

Notably, a high miscoverage rate does not always indicate that fewer cases are being sent for further testing. For example, at very high significance levels, most prediction sets are empty, which results in high miscoverage rates. However, in these cases, more samples are classified as ambiguous, leading to more referrals for further testing.

B. AMBIGUITY RATE ANALYSIS

The relationship between the ambiguity rates and significance levels exhibited a distinctive U-shaped pattern, as shown in Fig. 6(a). Initially, as the significance level increased to 0.25, ambiguity rates decreased. This trend could be attributed to the conformal predictor generating more singleton predictions at moderate significance levels. However, beyond 0.25, the ambiguity rates began to increase again, creating the observed U-shaped curve.

This behavior can be explained by examining two competing effects. Low significance levels lead to low miscoverage rates, which means the model is more likely to include both classes in the prediction set. This results in more ambiguous predictions. Conversely, high significance levels result in high miscoverage rates where the model tends to predict empty sets, also increasing the ambiguity.

While the minimum ambiguity rate occurs at approximately 0.25 significance level, suggesting an optimal operating point in terms of prediction precision, this setting may be unsuitable for clinical applications. A significance level of 0.25 indicates that only 75% of the prediction sets will contain the true class. This poses an unacceptable risk in medical diagnosis, as missing the correct HER2 status could lead to inappropriate treatment decisions.

It is important to note that this optimal point is not universal and depends on several factors, including the choice of the nonconformity measure, underlying feature distribution, and performance of the base model. Changes in any of these components can shift the location of the minimum ambiguity

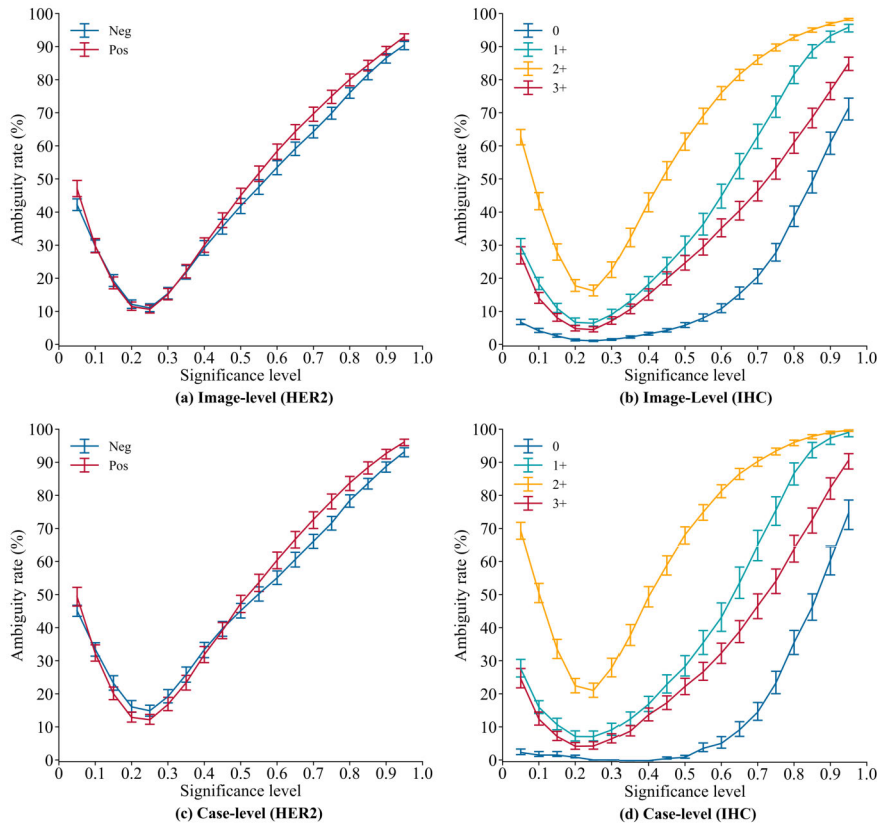


FIGURE 6. Ambiguity rates from the Random Forest classifier. Image-level results: HER2 status (a) and IHC score (b).Case-level results: HER2 status (c) and IHC score (d).

TABLE 3. Case-level ambiguity rates (%) with 95% CI (lower is better).

α	Model	HER2			IHC			
		Overall	Neg	Pos	0	1+	2+	3+
0.05	DT	68.5 (65.1-71.7)	65.7 (62.4-69.1)	72.0 (68.0-76.4)	41.2 (35.6-46.3)	57.5 (53.1-62.0)	80.3 (77.7-83.3)	62.1 (56.1-68.0)
	RF	47.0 (45.0-48.9)	45.2 (43.6-47.0)	49.3 (46.5-52.1)	2.3 (1.2-2.9)	27.6 (24.9-30.1)	69.5 (66.9-72.1)	24.5 (21.6-27.4)
	GBT	51.5 (49.4-53.7)	47.6 (45.9-49.2)	56.0 (52.5-59.2)	5.8 (3.7-7.2)	35.4 (32.5-38.3)	71.2 (69.0-73.5)	36.4 (31.8-40.5)
	XGB	47.4 (45.8-49.0)	46.8 (45.2-48.4)	48.8 (46.4-50.9)	3.6 (2.1-4.6)	33.7 (30.6-36.6)	69.8 (67.7-71.9)	23.0 (20.5-25.2)
0.10	DT	49.3 (46.4-52.2)	47.9 (44.6-50.7)	51.9 (48.0-55.5)	18.5 (14.3-22.1)	36.5 (32.3-40.4)	64.2 (61.1-67.3)	37.8 (32.8-42.3)
	RF	32.5 (30.5-34.5)	33.5 (31.5-35.5)	32.2 (29.7-34.6)	1.5 (0.4-1.9)	16.1 (13.9-17.9)	50.5 (47.6-53.4)	12.4 (10.2-14.1)
	GBT	35.7 (33.7-37.6)	34.6 (32.9-36.4)	37.3 (34.5-39.9)	1.6 (0.5-2.0)	19.7 (17.5-21.8)	53.0 (50.3-55.6)	20.1 (16.7-22.9)
	XGB	32.4 (30.4-34.1)	33.1 (31.2-35.1)	32.3 (30.0-34.5)	1.3 (0.2-1.5)	18.9 (16.2-21.3)	50.3 (47.5-53.1)	12.3 (10.5-13.9)

rate. Ideally, in clinical applications, we should aim for an optimal point at lower significance levels, closer to 0. This preference would minimize the miscoverage risk, ensuring more reliable predictions, although it may result in higher ambiguity rates.

The ambiguity rates across IHC scores exhibited a U-shaped distribution. As expected, equivocal cases demonstrated higher ambiguity rates than other IHC scores, as illustrated in Fig. 6(d). Table 3 presents the ambiguity rates for all the models at significance levels of 0.05 and 0.10.

We selected these significance levels because higher values would not be practical in clinical applications, as they would result in unacceptably high miscoverage rates. The analysis revealed that more sophisticated models (RF, GBT, and XGB) achieved lower ambiguity rates across all categories, with RF achieving the lowest ambiguity rate (47.4%) at a significance level of 0.05.

Although conformal prediction guarantees coverage, it does not directly control the ambiguity rate (predictions containing both classes or empty sets). However,

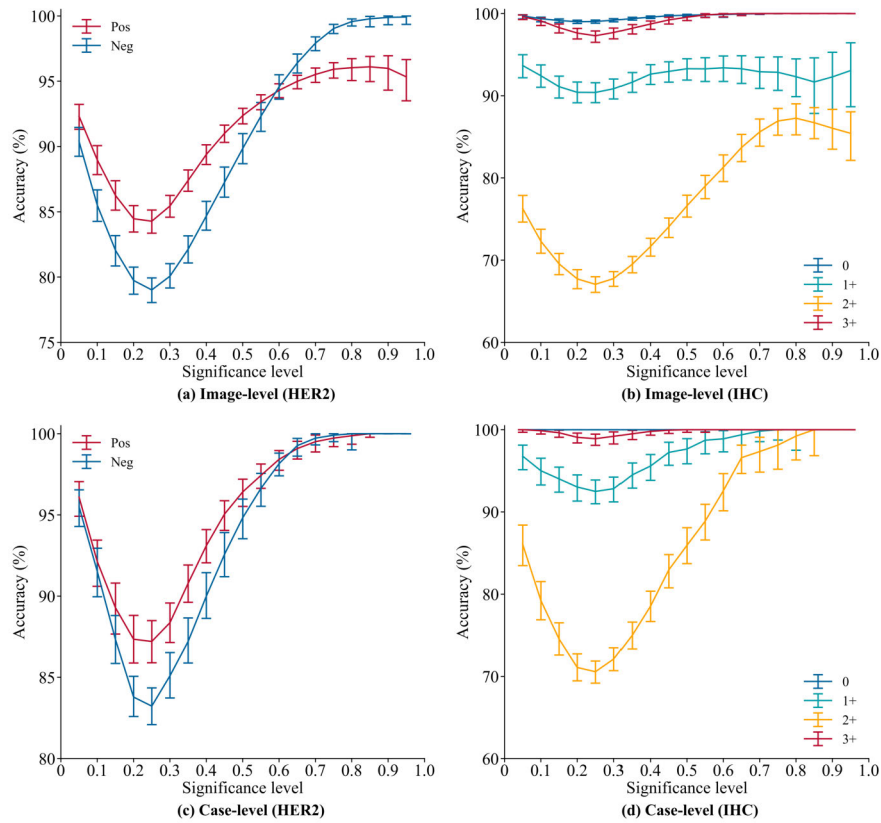


FIGURE 7. Accuracies across image and case levels, stratified by HER2 status and IHC score.

the expected ambiguity rate can be empirically estimated for given significance levels and data distributions. This allows pathologists to make informed decisions regarding additional testing requirements, balancing the need for further diagnostics against the confidence provided by the prediction sets.

C. ACCURACY ANALYSIS

The accuracies shown in Fig. 7 demonstrate U-shaped curves across the different significance levels. Similar to the ambiguity rates, the accuracies reached their lowest points at significance levels between 0.2 and 0.3. The shape of this curve can be divided into three regions:

At low significance levels, the empirical quantiles from the calibration set are very high, as shown in Fig. 8. This resulted in most prediction sets containing both classes, indicating ambiguous classifications. Under these conditions, singleton predictions occurred only for cases in which the model exhibited very high confidence, thus yielding accuracies despite the tendency toward ambiguity.

The middle region, which is indicated by the vertical dashed line in Fig. 8, represents the critical transition. At this point, the empirical quantile intersected the median of the nonconformity scores of the test sets at a significance level of approximately 0.25. Singleton predictions became more common, as the nonconformity scores from most test samples

did not need to be very high to avoid ambiguous predictions. However, forcing the model to make singleton predictions of HER2 status, especially in equivocal cases, resulted in low accuracy, because some cases were visually inseparable.

As the significance levels increased, the model became more selective in its predictions, owing to the decreasing empirical quantile values. This selectivity indicates that the test samples require high nonconformity scores to avoid empty prediction sets. Similar to the low-significance region, this selective behavior resulted in high accuracy, completing the U-shaped curve.

Table 4 shows the accuracies across the HER2 status and IHC scores. XGB achieved the highest overall accuracy compared with other models.

Table 5 presents the accuracies and reduction rates for the equivocal cases. The reduction rate, calculated as $(100 - \text{ambiguity rate})\%$, represents the proportion of cases with singleton predictions. The results showed that the accuracies for equivocal cases with negative HER2 status were lower than the positive status ($\sim 5\%$). This indicates that the models are more conservative in predicting positive cases, resulting in a higher false negative rate.

Among the evaluated models, RF achieved the highest reduction in equivocal cases (30.5%), whereas XGB demonstrated superior accuracy compared to all the models at a significance level of 0.05. The negligible difference in

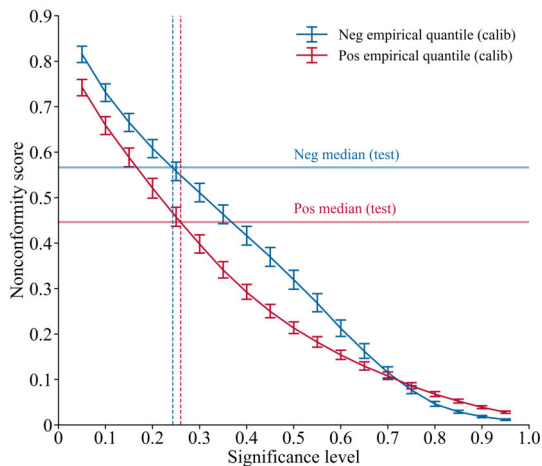


FIGURE 8. Empirical quantiles of HER2 status at image level derived from the Random Forest classifier. The median nonconformity scores are unaffected by the significance level as they are calculated from the test set while the empirical quantiles are calculated from the calibration set. The intersection point (dashed line) between empirical quantiles and median nonconformity scores identifies the region with highest singleton prediction frequency.

reduction ($\sim 0.4\%$) is compensated by XGB's significantly higher accuracy in both negative and positive HER2 status determination for equivocal cases.

In practical terms, with approximately 10 equivocal cases in our test set for each NCV, a 30% reduction achieved by XGB would decrease the number of cases requiring DISH testing from 10 to 7.

V. DISCUSSION

This study demonstrated the effective application of Conformal Prediction (CP) for uncertainty quantification in HER2 status prediction from IHC images. Our empirical results validate the theoretical properties of the CP framework by demonstrating reliable coverage guarantees across various significance levels. The framework exhibits a clear and interpretable trade-off between coverage and precision. Lower significance levels produce conservative predictions with higher coverage rates and larger prediction sets. Conversely, higher significance levels lead to more selective predictions with increased singleton predictions but elevated misclassification risks.

In medical applications, prioritizing low significance levels is crucial to minimize miscoverage, ensuring that patients receive further testing when necessary. While this conservative approach may result in more ambiguous predictions, it primarily impacts equivocal (2+) cases, as other categories (0, 1+, and 3+) are typically distinguishable through visual inspection. Although the accuracy tends to be higher at high significance levels, very low significance levels can also lead to increased accuracy. At very low significance levels, the model classifies only predictions with extremely high probability scores as non-ambiguous, resulting in higher prediction accuracy.

This observation aligns with the findings by Koopman et al. [7], who reported that reducing equivocal predictions inherently compromised the sensitivity of HER2 assessment. Notably, advanced classifiers such as Random Forest (RF) and XGBoost (XGB) achieved superior performance by maintaining high accuracy while generating more concise prediction sets. This suggests that model sophistication can effectively optimize the trade-off between accuracy and ambiguity in HER2 status prediction.

The integration of our CP framework into clinical practice offers several advantages for HER2 status assessment, particularly in managing equivocal cases which often present diagnostic challenges. At the case level with a significance level of 0.05, our XGB model achieved an overall accuracy of 90.1% (95% CI: 87.4-92.2%) for equivocal (2+) cases, with accuracies of 91.7% (95% CI: 87.3-94.8%) and 86.5% (95% CI: 82.0-90.4%) for correctly predicting HER2-positive and HER2-negative status respectively. The model particularly demonstrated strong performance compared to other tree-based classifiers, with boosting methods (XGB and GBT) outperforming the bagging approach (RF) in handling these challenging cases. Rasmussen et al. [16] evaluated the accuracy in predicting HER2 status for equivocal cases using ISH test results as the gold standard. Their deep learning model, trained on 115 equivocal cases, achieved 81% accuracy on their external validation set. While our model achieved higher accuracy, this comparison should be interpreted with caution due to key methodological differences: we trained on images from all IHC scores, predicted only low uncertainty cases, and had a smaller test set (10 cases versus their 36 cases). The lower accuracy of HER2-negative prediction in equivocal cases can be attributed to the class imbalance in our dataset, where 2+ cases with HER2-negative status (23 cases) were fewer than those with HER2-positive status (30 cases). Consequently, each misclassification of HER2-negative cases had a larger impact on the overall accuracy metrics. While there is no universal rule for selecting significance levels in clinical practice, lower values are preferred as they minimize the risk of missing true classes. The framework could be easily integrated into clinical workflows where pathologists capture images, specify a significance level based on their risk tolerance, and receive prediction sets to support their decision-making. These results suggest that our framework could assist pathologists in triaging equivocal cases, potentially reducing the number of cases that requires additional confirmatory testing while maintaining high diagnostic accuracy.

Compared to existing approaches, our results demonstrate both promising advances and limitations. Jakobsen et al. [6], using a Digital Image Analysis (DIA) system, achieved a 67% reduction in equivocal cases while maintaining 100% sensitivity. Our model achieved approximately 30% reduction in equivocal cases; however, direct comparison requires careful consideration because methodological differences. While we focused on evaluating non-ambiguous cases, the presence of false negatives in our predictions indicated

TABLE 4. Case-level accuracies (%) with 95% CI (higher is better).

		HER2				IHC			
α	Model	Overall	Neg	Pos	0 ^a	1+	2+	3+	
0.05	DT	92.8 (91.6-93.9)	94.1 (92.7-95.3)	91.7 (89.4-93.6)	100.0	90.4 (85.6-93.6)	83.6 (80.5-86.5)	99.9 (99.2-100.0)	
	RF	94.6 (93.6-95.4)	95.5 (94.3-96.6)	96.1 (94.9-97.0)	100.0	96.8 (95.1-98.1)	86.0 (83.5-88.3)	100.0 (99.7-100.0)	
	GBT	95.0 (94.1-95.9)	97.2 (96.1-98.0)	94.9 (93.7-96.1)	100.0	93.2 (91.1-95.0)	89.4 (87.3-91.3)	100.0 (99.6-100.0)	
	XGB	96.3 (95.4-97.0)	97.3 (96.2-98.2)	97.0 (96.1-97.8)	100.0	97.9 (96.2-99.1)	90.1 (87.4-92.2)	100.0 (99.2-100.0)	
0.10	DT	87.7 (86.6-88.8)	92.1 (90.9-93.2)	91.3 (90.0-92.4)	100.0	91.4 (89.0-93.4)	79.6 (77.0-81.9)	99.2 (96.8-100.0)	
	RF	90.3 (89.1-91.5)	91.6 (90.1-93.0)	92.2 (90.7-93.5)	100.0	95.0 (93.3-96.5)	79.3 (77.0-81.7)	99.9 (99.5-100.0)	
	GBT	90.3 (89.2-91.4)	92.4 (90.9-93.8)	91.4 (90.1-92.8)	100.0	91.2 (89.5-92.8)	80.9 (78.8-82.9)	99.8 (99.3-100.0)	
	XGB	91.4 (90.3-92.5)	92.4 (90.9-93.8)	93.1 (91.8-94.3)	100.0	96.0 (94.3-97.4)	81.8 (79.6-83.9)	99.8 (99.0-100.0)	

^aNo variation in measurements (100% consistent results).

TABLE 5. Case-level accuracies (%) for equivocal (2+) cases by HER2 status and reduction rates (%) with 95% CI.

α	Model	2+ Neg Acc.	2+ Pos Acc.	Reduction
0.05	DT	79.3 (73.0-85.3)	70.4 (72.8-84.9)	19.7 (16.7-22.3)
	RF	83.7 (78.8-88.1)	86.0 (80.8-90.0)	30.5 (27.9-33.1)
	GBT	81.3 (76.3-85.7)	91.7 (88.3-94.5)	28.8 (26.5-31.0)
	XGB	86.5 (82.0-90.4)	91.7 (87.3-94.8)	30.2 (28.1-32.3)
0.10	DT	74.9 (69.5-79.3)	78.4 (73.7-82.2)	13.2 (10.5-15.8)
	RF	74.4 (69.5-79.0)	82.8 (79.2-86.0)	24.1 (21.7-26.5)
	GBT	74.9 (70.0-79.3)	83.8 (80.6-86.7)	22.5 (20.3-24.7)
	XGB	78.7 (74.7-82.5)	84.8 (81.3-88.1)	23.8 (21.9-25.7)

sensitivity below 100%. Despite DIA's superior performance, our CP framework offers a complementary advantage: it could potentially be integrated with existing DIA systems by using their generated scores [5] as nonconformity measures, thereby adding uncertainty quantification capabilities to these established systems. Several factors contribute to this difference in the performance. The DIA analysis utilized whole-slide images, which provide more comprehensive information than our manually captured microscopic fields, potentially introducing selection bias and limiting the available information. In addition, the uniformity of the image brightness significantly affects model performance. DIA software also maintains consistent brightness and configuration across slides while microscope-captured images exhibit varying brightness levels, potentially leading to distribution shifts that affect model performance.

This study has several important limitations. The analysis was confined to a single dataset, which potentially limits the generalizability of our findings to diverse clinical contexts and patient populations. The lack of multi-center validation raises questions about the robustness of the model across different laboratory protocols and patient demographics. This is particularly important as CP relies on the exchangeability assumption between the calibration and test sets for valid coverage guarantees. The reliance on manually captured

microscopic fields introduces potential selection bias and variation in image quality, which may affect the model performance. Additionally, while the current approach using predetermined handcrafted features demonstrated effectiveness, it may not capture all relevant characteristics, particularly in equivocal cases where subtle patterns could be crucial for accurate classification.

Several promising directions for future research emerge from this study. Future research could focus on developing adaptive algorithms that automatically adjust significance levels based on case-specific factors, including tumor characteristics and patient history. This automation would enable more personalized, context-aware predictions while reducing the need for manual adjustments. Validation on larger, more diverse datasets is crucial, particularly through multi-center studies that would assess the framework's performance across different laboratory protocols and patient populations. Investigation of deep learning approaches or novel feature extraction methods could potentially reduce ambiguous cases and improve overall predictive performance. Of particular promise are the integration of attention mechanisms for automatic region-of-interest identification, self-supervised learning approaches to better utilize unlabeled data, and development of more robust features for equivocal cases.

The development of standardized protocols for model validation and uncertainty quantification in digital pathology would benefit the entire field. This includes establishing benchmarks for comparing different uncertainty quantification methods, developing guidelines for incorporating these tools into clinical workflows, and creating protocols for quality control and monitoring model performance in clinical settings. This research represents a significant step toward reliable automated HER2 assessment, demonstrating that principled uncertainty quantification can enhance the clinical utility of machine learning models in digital pathology.

VI. CONCLUSION

This study demonstrated the application of the CP framework for uncertainty quantification in HER2 status prediction from IHC images. By leveraging handcrafted features and

tree-based classifiers, our approach introduces a systematic method for handling the prediction uncertainty in digital pathology. A key advantage of the proposed framework is its ability to identify and appropriately handle cases with high uncertainty, particularly equivocal cases, which may require additional testing. Our empirical analysis of the miscoverage rates, ambiguity rates, and accuracies at both the image and case levels provides valuable insights into the behavior of the model. Notably, our findings align with clinical observations, demonstrating higher ambiguity rates and lower accuracies for equivocal cases, thus validating the ability of the framework to capture clinically relevant uncertainty patterns.

The proposed approach represents a step toward more reliable and transparent automated HER2 status assessment systems, in which uncertainty quantification plays a crucial role in supporting clinical decision-making. Although there is room for improvement compared to existing DIA solutions, our framework establishes a foundation for developing more sophisticated uncertainty-aware systems in digital pathology.

APPENDIX A HARALICK FEATURES

We used the implementation from the *scikit-image* (v.0.25.1) [39] library. Let $P_{i,j}$ denote the normalized element of the GLCM $_{N \times N}$ with (r, θ) where i and j denote the horizontal and vertical indices, respectively. The seven Haralick features are defined as follows:

$$\text{Contrast} = \sum_{i,j=0}^{N-1} P_{i,j} \cdot (i - j)^2, \quad (8)$$

$$\text{Dissimilarity} = \sum_{i,j=0}^{N-1} P_{i,j} \cdot |i - j|, \quad (9)$$

$$\text{Homogeneity} = \sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1 + (i - j)^2}, \quad (10)$$

$$\text{ASM} = \sum_{i,j=0}^{N-1} P_{i,j}^2, \quad (11)$$

$$\text{Energy} = \sqrt{\text{ASM}}, \quad (12)$$

$$\text{Entropy} = - \sum_{i,j=0}^{N-1} P_{i,j} \cdot \ln(P_{i,j}), \quad (13)$$

$$\text{Correlation} = \sum_{i,j=0}^{N-1} P_{i,j} \cdot \frac{(i - \mu_x)(j - \mu_y)}{\sigma_x \sigma_y}, \quad (14)$$

where

$$\mu_x = \sum_{i,j=0}^{N-1} i \cdot P_{i,j}, \quad (15)$$

$$\mu_y = \sum_{i,j=0}^{N-1} j \cdot P_{i,j}, \quad (16)$$

TABLE 6. Hyperparameter search space for each model.

Model	Hyperparameter	Minimum	Maximum
DT	max_depth	3	20
	min_samples_split	2	20
	min_samples_leaf	1	20
RF	n_estimators	10	200
	max_depth	3	20
	min_samples_split	2	20
	min_samples_leaf	1	20
GBT	n_estimators	10	200
	max_depth	3	20
	min_samples_split	2	20
	min_samples_leaf	1	20
	learning_rate	0.001	0.1
XGB	n_estimators	10	200
	max_depth	3	20
	min_child_weight	1	20
	subsample	0.5	1.0
	colsample_bytree	0.5	1.0
	learning_rate	0.001	0.1
	gamma	0	5

$$\sigma_x = \sqrt{\sum_{i,j=0}^{N-1} P_{i,j} \cdot (i - \mu_x)^2}, \quad (17)$$

$$\sigma_y = \sqrt{\sum_{i,j=0}^{N-1} P_{i,j} \cdot (j - \mu_y)^2}. \quad (18)$$

APPENDIX B HYPERPARAMETERS

In our experiments, we used models from the *scikit-learn* (v.1.5.1) [40] library, except for the XGB classifier, which we used the *xgboost* (v.2.1.2) [41] library. Because the models has many hyperparameters, only the hyperparameters that we optimized are listed in Table 6. The remaining hyperparameters were set as default values. We used the Tree-structured Parzen Estimator implemented by Optuna [42] with 50 trials per hyperparameter optimization. The objective of the validation set was to maximize the area under the receiver operating characteristic curve (AUC).

ACKNOWLEDGMENT

The authors express their sincere gratitude to all the patients who contributed tissue samples to this study. Their participation has been invaluable in advancing their understanding of breast cancer diagnosis. They also thank the pathologists and staff at King Chulalongkorn Memorial Hospital for their assistance in image acquisition and annotation.

The language and readability of this manuscript were improved using Claude (Anthropic), Paperpal Preflight, and Grammarly writing assistance tools.

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